Cape Peninsula
University of Technology

INDUSTRIAL COMPUTER DESIGN PROJECT 3
DES371S - 2026

(GA9 + Broadly Defined Problem Alignment)

Title of Project: NLP Based Chatbot System

Bachelor of Engineering Technology: Computer Engineering
Department of Electrical, Electronic and Computer Engineering
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1.PROBLEM OVERVIEW

1.1

Why your problem is NOT simple.	The project is not straightforward because it integrates conversational AI, generative AI, voice interaction, multilingual communication, emotion recognition, reminder management, weather integration, and database storage into a single system. In addition to storing user data and producing intelligent responses, the chatbot must process text, voice, images , and camera input.
What makes it complex or multi-dimensional? This is a broadly defined feature.	The project contains many interacting subsystems including NLP, speech recognition, emotion analysis, database management, multilingual processing, APIs, and generative AI. It must balance usability, privacy, performance, accessibility, and academic integrity while handling uncertain user interactions and various implementation strategies.
What decisions do you need to make as a technologist?	Decisions include selecting AI models, choosing speech recognition technologies, deciding how multilingual support will function, determining how emotion detection will be implemented ethically, designing the database structure, integrating APIs, and balancing trade-offs between system speed, accuracy, privacy, usability, complexity, and computing resources. When determining how sophisticated features like voice interaction, generative AI, and emotion recognition will function well without adversely affecting system efficiency or user experience, trade-offs must also be taken into account.

1.2

What is the problem?	Many students struggle to access personalized academic support, manage study schedules, stay motivated, and receive immediate assistance when facing academic or emotional challenges. Students fail due to lack of understanding and limited academic resources such as past question papers. Many face difficulties in organising study schedules, handling stress, accessing timely help when struggling academically or emotionally, and obtaining individualised academic support. Lack of understanding and academic resources contribute to students' academic failure. Language barriers, stress during exams and assessments, time management, and revision preparation are other challenges that students face. Many students find it challenging to get ongoing academic support because private tutors are sometimes costly and not available all the time.
Who is affected?	Students who require academic guidance, study support, multilingual communication, emotional support, reminders for assignments and exams, note summarisation, quiz creation, and flexible access to learning assistance through text and voice interaction are primarily impacted.
Why is it important?	Poor academic support, stress, lack of motivation, inefficient study techniques, language obstacles, and poor time management can all have a detrimental impact on students' academic achievement, confidence, mental health, and general success. This makes the issue crucial. Additionally, students struggle to obtain ongoing, reasonably priced learning assistance. Addressing these challenges can improve learning accessibility, student engagement, academic preparedness, and educational outcomes through personalized and interactive academic support.

2.LITERATURE REVIEW

2.1 INTRODUCTION TO THE PROBLEM DOMAIN

Many people experience challenges in accessing quick and reliable support services. In education students may fail due to lack of understanding, limited academic resources, and insufficient access to chapter and exam style questions. Due to hospital overcrowding, patients may have to wait a long time or be turned away. Similarly, customers at shopping centres often experience long queues and delayed assistance. These challenges created the need for systems that can provide fast, accessible, and continuous support.

To address these issues, chatbot systems were developed. Chatbots are computer programs designed to mimic human conversation and assist users by responding to enquiries, providing information, and carrying out tasks automatically (Church & Finn, 2025). Over time, chatbot systems have evolved from simple retrieval-based systems to more advanced conversational AI, voice, generative AI, and hybrid chatbots that can understand natural language, generate new content, and support personalized interactions.

This literature review discusses different types of chatbots, their technologies, advantages, limitations, and suitability. It also justifies the selection of a hybrid conversational AI chatbot with generative capabilities and voice interaction support for the proposed student support system.

2.2 EXISTING APPROACHES

Retrieval Based Chatbots

Retrieval-based chatbots identify and retrieve answers that are already written and stored in a database based on user's questions (Callin.io, 2025). To evaluate user questions and match them with existing responses, these systems mostly rely on pattern-matching algorithms. Pattern-matching algorithm is a process of searching for a specific sequence of characters such as matching words, phrases or text patterns within a long text and finding information that looks similar to the sequence in the database. Retrieval based chatbots also use rule-based frameworks where the chatbot follows predefined rules to answer user questions. For example if the user types a "Hi "message chatbot replies with "Hello". These systems produce accurate responses, work well for frequently asked questions (FAQs) and are used in closed domain scenarios where they are trained for a specific area of knowledge and can only answer questions related to that area (Codecademy, n.d.). However, retrieval -based chatbots only use stored information and cannot handle unseen and new cases if predefined responses do not exist (Codecademy, n.d.).

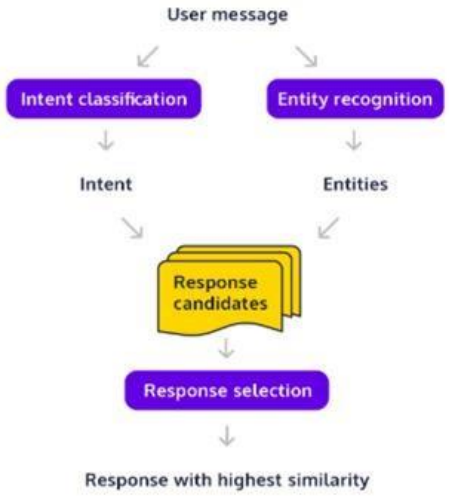


Figure1 illustrates how a retrieval-based chatbot operates. The user message is first analysed through intent classification and entity recognition to determine the user's request and match it with stored responses. The system then generates possible responses (response candidates) from a predefined database of stored responses and selects the response with the highest similarity to the user query (Callin.io, 2025).

Conversational AI Chatbots

Unlike retrieval based chatbots that support only predefined questions and answer options, conversational AI chatbots can understand user's questions regardless of how they are expressed (Church & Finn, 2025). Conversational AI powered Chatbots use natural language processing, a method of making the machine or a computer understand the human language, and artificial intelligence (AI) to comprehend user intent and provide responses. Conversational chatbots can understand different sentence structures and provide more intelligent responses. The machine learning algorithms supporting AI chatbots enable them to self- learn and develop an increasingly intelligent knowledge base of questions and responses that are based on user interactions. Conversational AI Chatbots are no longer limited to responding to frequently asked questions. They can perform a wide range of tasks, including scheduling meetings, sending messages to customers, assisting in completing sales, and making medical appointments, among other functions.

In addition,conversational chatbots can recall previous user interactions and apply this information in future conversations (Church & Finn, 2025). For example, a chatbot may use a student's name in its responses if it was provided during an earlier interaction, thereby creating a more personalized experience. Similarly, in an online shopping context, such as with Checkers, a chatbot can recognize returning customers and use stored information, such as delivery addresses and payment details, to streamline and complete the ordering process more efficiently.

Despite their capabilities conversational AI chatbots have several limitations. Conversational chatbots are more complex to develop, require significant computing resources and often rely on large datasets for training (Sherman, 2026). In addition, they require continuous maintenance and updates, including retraining with new data and adapting to changes in processes or information. They fail in creativity which requires out of the box thinking. For example they cannot generate images. There are also concerns related to security and data privacy, as sensitive or private information entered into the chatbot systems may get exposed or shared without the user's permission sometimes leading to violation of organizational policies (Hassan, N, 2025). Additionally, when there is insufficient or poor-quality input data, conversational AI

chatbots may produce inaccurate or irrelevant responses. Furthermore these chatbots still struggles with emotional detection.

Voice Chatbots

Another chatbot type is a voice chatbot, which enables users to communicate with the bot by speaking through a microphone to it instead of typing (Church & Finn, 2025). AI powered voice chatbots provide the same sophisticated capabilities as conversational AI chatbots, but they operate through voice-based platforms. Voice chatbots rely on speech to text and text to speech technologies to convert spoken words to text and text to spoken words. By combining Natural Language Processing with computer and telephony systems, these chatbots can interpret spoken queries, assess user needs, and deliver accurate, conversational responses. This is beneficial because it encourages multitasking unlike conversational AI chatbots, for example, a student may be using a voice chatbot to comprehend lecture content while cooking or organising clothing (An, 2025). Additionally, voice chatbots play a huge role in customer support where they provide quick responses to questions reducing customer wait times and also benefiting businesses by reducing the need to employ many employees. These chatbots also excel in Airline Operation
Department where they provide real time updates on flight schedules, delays and changes.

Voice chatbots may struggle to accurately interpret user input due to variations in accents and may ask users to repeat questions or generate irrelevant responses (An, 2025). The precision of voice chatbots can also be greatly impacted by background noise where the chatbot can misunderstand user questions if users are in noisy environments. There are privacy issues with voice interaction technology. Chatbots that use Voice-Use Interfaces (VUIs) where the user does not have to press the microphone button when speaking frequently use always-listening systems, which may unintentionally record private conversations. This raises questions about the usage and storage of user's personal data. Additionally, the chatbot requires a strong internet connection and may delay in answering question or not answer at all when there is poor connection.

Generative AI chatbots

Generative AI chatbot is a type of chatbot that can create new content using large language models (LLMs) (CSG, 2025). LLMs analyse existing content to find patterns and make predictions, enabling them to create new text, images, computer code and even music in response to user inputs unlike Conversational AI chatbots which struggle to generate images. These chatbots are also capable of recognizing, summarizing, translating user input and generating exam style questions for students based on uploaded notes. For example, they can summarise student notes and generate flash cards or create simple diagrams to support understanding of complex concepts. Generative AI chatbots act as tutors providing immediate support to students and creating a more engaging learning environment (Aldhafeeri, et al., 2025).

Some businesses develop generative AI chatbots to handle new risks and ensure safe implementation of AI systems with the help of LLM security, Cyber Incident Response, and Security Log Summarization (Aldhafeeri, et al., 2025). These applications prevent the chatbot from being easily hacked by finding weaknesses, reduce risks and improve the chatbot's overall level of safety and protection from cyber threats. In online shopping generative chatbots go beyond assisting customers with product inquiries and personalised recommendations, they also enable customers to track their order.

Even though generative chatbots have shown many successes they also have limitations that affect their effectiveness. One of these limitations is the generation of hallucinations in which the chatbot gives incorrect information that sounds fluent and intelligent resulting in reduced

customer trust (University of Leeds, n.d.). Generative chatbots may also use information from existing books, articles and journals without the permission of original authors when responding to student question, sometimes without giving credit or references. This may lead students to be punished for copyright and ownership or plagiarism (University of Leeds, n.d.).Another potential concern is the over reliance on generative AI chatbots that may reduce independent thinking and problem-solving skills, as users may become overly dependent on the system to complete tasks and assignments instead of actively engaging with the learning material (Stike, 2025).

Hybrid Chatbots

A hybrid chatbot is an intelligent system that effectively combines AI features such as Natural language processing and machine learning with rule-based techniques used in retrieval based chatbots (Church & Finn, 2025). Rather than relying solely on machine learning, stored information or predefined scripts, it integrates both approaches. This enables the chatbot to handle a wider range of queries and provide more accurate responses. As a result, it offers a more flexible and useful experience for users. Another advantage is that it can handle different ways of asking questions, not just one fixed format, while still giving correct and reliable answers to common questions and can forward complex or sensitive issues to human agents when necessary. This enhances the overall user experience. Furthermore, hybrid chatbots are easier to maintain and update. The rule-based component can be modified as needed, while the AI component continues to learn and improve over time. As a result, the system becomes more efficient and scalable.

Although Hybrid chatbots combine the benefits of retrieval, Natural language processing and machine learning they still have their limitations. Hybrid chatbots may struggle to deal with complex problems that requires out of the box thinking and creativity unlike Generative chatbots (Church & Finn, 2025).

2.3 Comparative Analysis Table

Method	Technologies	Advantages	Limitations	Suitability
Retrieval Based	Pattern matching algorithm and Rule based frameworks	Accurate responses. Good for FAQs	Rely on stored responses. Cannot handle unseen or new cases	Closed domain scenarios

Conversational AI Chatbots	NLP, AI, ML,	Understand different user questions; intent recognition; personalised responses; remembers previous interactions; handles complex tasks	Can produce inaccurate or irrelevant responses, Possible exposure of sensitive information, Complex to develop, continuous maintenance and retraining, limited creativity	Customer support, Online shopping, education and health care
Voice Chatbots	NLP, AI, Speech-toText, Text-toSpeech, VUIs, Telephony systems	Hand free interaction, encourages multitasking, reduce customer wait times, Reduce the need to employ many employees	Difficulty understanding accents, affected by background noise, Privacy concerns, Requires strong internet connection	Airline Operation departments, customer support, educational assistant, smart devices
Generative AI chatbots	AI, NLP, ML, LLMs, LLM security systems, Cyber Incident Response systems, Security Log Summarization systems	Creates new content. High creativity, Identify weaknesses in AI systems. Protect against cyber threats. Enables order tracking	Generates hallucinations. May use information without permission from original authors May reduce independent thinking and problem-solving skills on users.	Education Support, cybersecurity applications and online shopping

Hybrid Chatbots	Rule based frameworks, AI, NLP and ML	Gives reliable answers to FAQs, Provides more accurate responses, efficient and scalable, easier to maintain and update.	Limited creativity, struggles with complex problems	Businesses ,customer support ,banking

4.APPROACHE JUSTIFICATION

In light of what has been discussed above, the method chosen for this project is a hybrid conversational AI chatbot with generative capabilities and voice interaction support. This decision is driven by the specific goals and functionalities of the chatbot as a comprehensive student support tool. The chatbot is not limited to a single function; instead, it is designed to handle a variety of academic and personal support tasks, which require flexibility, adaptability, and intelligent response generation.

Firstly, the conversational AI component is essential because the system must understand and respond to a wide range of user queries expressed in natural language. Students may ask questions in different ways, and the chatbot must accurately interpret user intent in order to provide relevant and meaningful assistance. This is particularly important for answering academic questions, offering study tips, engaging in small talk, and maintaining a natural conversational flow.

Secondly, generative AI capabilities are necessary because the system goes beyond simply retrieving predefined responses. For example, the chatbot will analyse uploaded study notes, generate quiz questions, summarise content, and provide explanations. These tasks require the ability to create new, context aware responses rather than selecting from a fixed database. In addition, the system will generate and display real-time weather information visually, such as showing icons representing conditions such as sunshine, clouds, or rain, along with temperature values. This allows students to quickly understand the weather conditions while interacting with the chatbot. Generative functionality also supports personalized motivational messages and adaptive learning interactions, which enhance student engagement.

The hybrid aspect of the system is further justified by the inclusion of structured features such as scheduling assignment, test, and exam reminders, as well as storing user profiles, history and preferences. These functions rely on retrieving and managing stored data efficiently, which aligns with retrieval-based approaches. By combining retrieval mechanisms with generative AI, the system ensures both reliability for structured tasks and flexibility for open-ended interactions.

Additionally, voice interaction support has been included to improve accessibility and convenience. Students may not always be able to type, especially when multitasking, so allowing voice input and output enables more natural and efficient communication. The ability of the chatbot to read uploaded notes aloud further enhances its usefulness as a study companion.

Overall, the choice of a hybrid system with generative capabilities and voice interaction support ensures that the chatbot can deliver a well-rounded, personalized, and interactive learning experience. This design allows the system to effectively meet the diverse needs of students, ranging from academic assistance and organization to motivation and accessibility.

To prevent misuse and over-reliance on the system, specific limitations have been incorporated into the chatbot design. The system is intended to support learning through explanations, study guidance, and academic assistance rather than generating complete assignments or direct submission answers. When users request complete solutions, the chatbot will instead provide structured guidance and explanatory support. This approach promotes academic integrity and encourages independent learning and critical thinking among students.

The proposed system will also incorporate emotion-aware interaction features aimed at improving student support and engagement. With user consent, the chatbot will access the device camera to analyse facial expressions and emotional states, particularly during academically stressful situations such as tests and examinations. Based on the detected emotional state, the system will automatically provide motivational, encouraging, and growth-oriented messages intended to reduce stress, improve confidence, and create a more supportive learning environment. Ethical and privacy considerations will be strictly maintained by ensuring that camera access is only granted following explicit user permission.

The system will further support multilingual communication, allowing students to interact using their preferred language through both text and voice input. Supported languages will include English, IsiXhosa, and IsiZulu. Speech recognition and natural language processing technologies will enable the chatbot to understand and respond appropriately across these languages, thereby improving accessibility, inclusiveness, and user engagement for a broader range of students.

Finally, the chatbot will include a feedback mechanism through which students may evaluate their experience and provide suggestions for improvement. The collected feedback will be utilised to assess system performance and identify areas requiring enhancement during future stages of development.

5.SYSTEM OVERVIEW DIAGRAM

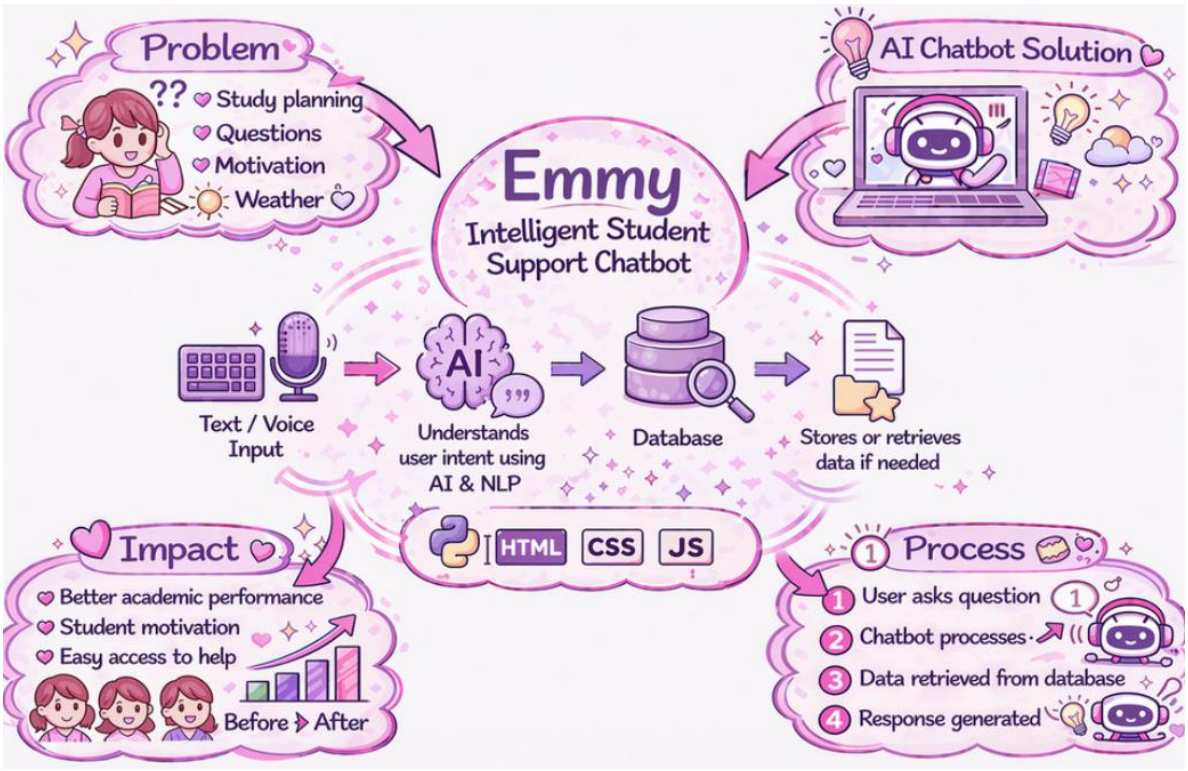


Figure 2: System overview

6. SYSTEM STAGES AND OUTPUTS

Stage	Description	Outputs
Stage 1	Using a web-based chatbot platform, the system gathers user input via text, audio, submitted images, and camera interaction.	Text, voice, image uploads, and camera input with permission management are all accepted by this useful chatbot interface.
Stage 2	NLP, speech recognition, multilingual support, and emotion detection are used by the system to process user input.	Personalised motivational replies, multilingual communication, accurate intent detection, and emotion recognition.
Stage 3	The system generates study materials, schedules, reminders, summaries, and weather updates by combining generative AI and retrieval features.	Created quizzes, summaries, study tips, calendars, reminders, history monitoring, and realtime weather data.
Stage 4	The system offers growth mentality and motivation while enforcing academic integrity and emotional support regulations.	Responses from chatbots that are safe, supervised, and help learning without directly answering assignments.

Stage 5	The system responds through text, audio, notifications, images, and multilingual output while collecting user feedback.	Reminders, motivational pop-ups, voice playback, multilingual chatbot responses, and saved user reviews.
Stage 6	The complete chatbot system will be tested for functionality, AI accuracy, multilingual support, emotion detection, voice interaction, database operations, usability, and performance before deployment.	A stable, secure, accurate, and fully tested AI student support chatbot system.

6.1 Stage Explanation

Stage	Stage Explanation	Tools/Software	How will success be measured
Stage1	Text, speech, uploaded images, and camera interactivity will all be used by the system to gather user input. Students will use a web-based interface to communicate with the chatbot. Before using the camera to assess emotions and enhance individualised student support, the system will ask for permission.	React.js for the chatbot interface. HTML, CSS, and JavaScript for frontend and user interaction. Web Camera API to access the camera stream and request permission. GitHub for source code management and collaboration, and Visual Studio Code for development and testing.	Success is determined by how well text, voice, uploads, and camera permissions are handled without mistakes or delays.
Stage2	System processes input using NLP, speech recognition, multilingual support, and emotional detection, to identify intent and emotional state,	Python for main backend language. SpeechRecognition to convert speech to text. OpenAI API for conversational AI and personalised motivational responses. OpenCV and DeepFace to analyse emotions. spaCy for NLP and intent recognition, and multilingual NLP libraries, open-source translation tools, language datasets will support IsiZulu, IsiXhosa, and English.	Evaluated by the relevance of motivating responses, multilingual output, emotion recognition, and intent detection accuracy.
Stage3	combines generative AI with retrieval to create summaries, tests, study advice, and inspirational messages while accessing schedules, reminders, and history. Also shows the current weather.	MySQL for storage (profiles, history, schedules and reminder). Flask will connect the frontend, database, and AI services. OpenAI API for content generation. OpenWeather API for weather data.	Determined by the created content's correctness, relevance, and speed of retrieval.
Stage4	applies decision making guidelines for both emotional support and academic integrity. Assignment assistance necessitates student work,	Flask validation will guarantee guided support rather than direct responses; OpenAI moderation tools will identify misuse or inappropriate	Determined by how well the rules are enforced, how well the support is provided, and how

	and motivational and growth mindset messages will appear while learning.	requests; and Python rule-based logic will handle academic integrity, motivating triggers, and emotional support.	direct assignment answers are avoided.
Stage5	provides answers through multilingual output if requested, text, audio, notifications, and images. includes feedback gathering and inspirational pop-ups.	Google Text-to-Speech (gTTS) will convert responses and notes into speech. React.js UI components and weather icon libraries will display weather and motivational pop-ups. Feedback forms connected to MySQL will store user ratings and suggestions. Notification modules will send reminders for assignments, tests, and exams	Success will be measured through successful voice playback, correct visual display, response accuracy, and positive user interaction experience.
Stage6	Before being deployed, the chatbot system will be tested for functionality, AI correctness, multilingualism, emotion detection, database, usability, and performance.	Selenium will test the web interface and user interaction. PyTest will test backend functions and Flask routes. Postman will test APIs. MySQL Workbench will test database operations. Apache JMeter will test performance and response speed. SpeechRecognition and gTTS will test voice input and output. OpenCV and DeepFace will test emotion recognition. spaCy will test NLP and intent recognition. Browser Developer Tools will test permissions and frontend errors. GitHub will support bug tracking and version control. SpeechRecognition and gTTS will test voice input and output. OpenCV and DeepFace will test emotion recognition. spaCy will test NLP and intent recognition. Browser Developer Tools will test permissions and frontend errors. GitHub will support bug tracking and version control.	Reduced errors, precise AI and multilingual responses, reliable system performance, accurate emotion detection, appropriate database functionality, successful voice interaction, and positive user feedback will all be used to measure success.

7.PROJECT PLAN

Task	Duration	Task Type
Topic Selection and Approval	3-8 March	Sequential
Research and Literature Review	1-13 April	Parallel
Requirements Gathering	13-18April	Parallel
System Analysis and Planning	18-27 April	Sequential
Proposal Writing	1-9 May	Sequential
Proposal Submission	17 May	Sequential
Database Design	June - Week 2 to Week 3	Parallel
User Interface Design	June - Week 3 to Week 4	Parallel
Chatbot Development	July -Week 1 to Week 3	Sequential
AI Feature Integration	July- Week 3 to Week 4	Parallel
Voice Interaction Integration	July - Week 4 to Week 5	Parallel
Testing and Debugging	August- Week 1 to Week2	Sequential
Final Report Writing	August- Week 3 to Week 4	Parallel
Final Report Submission	First Week October (according to subject guide)	Sequential
Presentation	First Week October (according to subject guide)	Sequential

Project Milestones

- Proposal Submission – 17 May
- Chatbot Development Completed – End of July
- Testing Completed – August Week 2
- Final Report Submission – First Week of October
- Final Presentation – First Week of October

8. PROJECT BUDGET

Most tool, frameworks, and libraries used in the project are open-source and freely available.

Item / Resource	Purpose	Estimated Cost
Laptop / Computer	System development, testing, and documentation	Existing Resource
Internet / Wi-Fi	Research, development, API access, and testing	Existing Resource
Visual Studio Code	Software development environment	Free
React.js	Frontend development	Free
Flask Framework	Backend development	Free
MySQL Database	Data storage and management	Free
GitHub	Version control and collaboration	Free
OpenCV Library	Emotion detection implementation	Free
DeepFace Library	Facial emotion recognition	Free
SpeechRecognition Library	Voice input processing	Free
Google Text-to-Speech (gTTS)	Voice output functionality	Free
OpenWeather API (Free Tier)	Real-time weather information	Free
Multilingual NLP Tools	IsiZulu, IsiXhosa, and English support	Free
Selenium Testing Tool	User interface testing	Free
PyTest	Backend testing	Free
Postman	API testing	Free
Apache JMeter	Performance testing	Free
Browser Developer Tools	Frontend debugging and testing	Free
Documentation and Printing	Printing the final report and documents	R100

Estimated Total Cost:R100

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